FACULTY OF ENGINEERING AND TECHNOLOGY

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BIS 3218 BUSINESS INTELLIGENCE SYSTEMS **DEADLINE: 8 JAN 2026**

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ASSIGNMENT 1: INSTRUCTIONS TO CANDIDATES

This is a group assignment of 6 members.

50% of your Final Assessment.

IMPORTANT **The University requires students to adhere to submission deadlines for any form of assessment. Penalties are applied in relation to unauthorized late submission of work.**
- **Coursework submitted after the deadline but within 1 week will be accepted for a maximum mark of 50%.** - **Work handed in following the extension of 1 week after the original deadline will be regarded as a non-submission and marked zero.**
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Academic Honesty Acknowledgement

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1.0 Case Study Development

1.1 Problem Identification

The current business environment heavily relied on data-driven decision-making to improve their operational efficiency and strategic planning. However, there are some businesses still struggle in analyse their historical data to predict future demand, which lead them unable to make a good decision. The core business problem addressed in this project is the lack of short-term retail sales performance that limits management's ability to identify the current market trend and risk early to allow them allocate resources optimally. Without predictive analytics, the organization may face issues in inefficient operational planning and missed the opportunities for performance improvement. By applying predictive modelling techniques, the organization can generate sales forecasts that enables management to make more informed strategic and operational decisions, improve resource allocation, and enhance overall business performance.

1.2 Dataset Selection

To address the business problem, our project had chosen a superstore sales dataset that contains four years retail sales data. The dataset reflects a realistic retail sales environment and suitable for predicting as past sales patterns are used to predict future demand. The dataset includes sales-related variables that can be used to build predictive models for short-term sales forecasting. However, it contains raw and unprocessed data as the dataset originates from a real-world business context. Therefore, data preprocessing is required before the dataset can be effectively used for predictive modelling. After preprocessing, the cleaned dataset will become input for building predictive models aimed at forecasting retail sales performance.

1.3 Business Context and Goals

Accurate sales forecasts play a important role in supporting an effective decision-making related to inventory management, planning, promotion strategies and financial performance. Based on the historical reports provide limit information for an organization to make a good decision. Thus, predictive analytics enable businesses to understand future demand and respond to it. The goal of this project is to support the business by developing a predictive model that forecast retail sales based on their previous data.

The specific goal for this project is:

1. To develop two predictive models (random forest and gradient boosting) that forecast sales.
2. To compare which model is the best for predicting by measure their performance using appropriate metrics.
3. To find the most influential variable that affect the prediction.

By achieving these goals, the project will show how predictive analytics can use historical data turn into an actionable strategy that enable organizations to improve forecasting accuracy, optimize resource allocation, and make informed, proactive business decisions.

1.4 Literature Review

1.4.1 Predictive Analytics in Retail Sales Forecasting

Retail businesses are operated in environments where demand variability is high, product life cycles are low, and seasonal and promotional impacts are high. Retailers are producing vast amounts of historical sales data, although traditional descriptive reporting is constrained in its ability to explain past performance and is not much help in predicting future demand. This disparity is especially important regarding short-term decision-making situations, where faulty predictions may cause inventory shortages, overstocking, redundant staffing, and promotional plan destabilisation.

Predictive analytics is stressed as an essential development of reactive to proactive control of the retail process as consistently noted in the literature. Predictive analytics helps retailers make more precise estimates regarding future demand by utilizing the history of sales data and sophisticated analytical tools to synchronize the decisions made by the operation division with the forecast. The accuracy of forecasting is consistently cited as a prerequisite of continuous retail operations because the errors get transmitted throughout the supply chain, marketing, and financial planning processes (Eglite & Birzniece, 2022).

A systematic review of the retail sales forecasting literature shows that, predictive models are generally more effective in dealing with complex sales patterns compared to traditional linear and statistical models. Deep learning and machine learning algorithms are especially useful in non-linear relationships, changing trends, and seasonality that are embedded in retail time-series information. In various retail scenarios like grocery, apparel, e-commerce and shopping malls, highly signalized predictive models tend to achieve lower forecast errors compared to techniques such as ARIMA and exponential smoothing particularly when the sales data are volatile and non-periodic demand (Eglite & Birzniece, 2022).

In addition to model performance, predictive analytics is strategically important in enhancing business decision making. The results of forecasts have been extensively applied to aid in inventory planning, demand allocation, and the scheduling of resources. Precise short-term predictions lower the chances of overstocking and stockouts, assisting retailers in maintaining the service levels and minimising holding and operating costs. Predictive models also can elevate the effect and timing of promotions by projecting anticipated demand surges and declines, instead of relying on them to react once the situation is destabilized (Magrini, 2024).

Following up , empirical research findings are also recent and indicate that the incorporation of other variables that affect sales enhances the accuracy of forecasting. Retail demand is not only influenced by past sales, but also by the contextual factors including holidays, salary cycles, weather condition, and special events. Predictive models that combine these variables

have better performance in that they are more realistic in representing the actual consumer behaviour. The application of both temporal learning and contextual feature extraction in hybrid neural network models have shown improved performance, demonstrating the relevance of multivariate forecasting methods in retail settings (Mansur et al., 2025).

Besides, the comparative studies of machine learning models indicate that no single traditional model is sufficient on its own to support the complexity of retail demand patterns. Even though the linear and statistical approach can still be used in stable settings, they fail to adapt to non-stationary data and sudden shifts in demand. Ensemble methods and neural networks are machine learning models that are better suited to the evolving consumer behaviour and changing patterns. Deep and hybrid architectures have a constant better accuracy, particularly in the short-term forecasting horizons, which are sensitive to operational planning (Mustapha & Sithole, 2025).

After that, even more recent developments are aimed at trade-offs between forecast accuracy and model efficiency and stability. Deep neural networks have a great predictive power, but their complexity may cause computational and interpretability difficulties. To deal with this, more recent forecasting methods investigate simplification of models without compromise to accuracy. Comparative analysis has demonstrated that custom-designed neural architectures are able to deliver high forecasting capabilities and can be scaled to use in real-world retail applications, especially when short-term sales forecasting is needed (Theodoridis & Tsadiras, 2025).

All in all, the literature confirms that predictive analytics has become a necessary component of contemporary retail sales forecasting. Predictive models help retailers avoid operational waste, predict changes in demand, and optimise inventory and resources using historical sales data converted into future insights. The predictive analytics approach has broadly been demonstrated to be an essential part of making data-driven decisions in retail in both scholarly and practical settings, notably in short-term forecasting scenarios where accuracy has a direct effect on a business (Eglite & Birzniece, 2022; Magrini, 2024; Mansur et al., 2025; Mustapha & Sithole, 2025; Theodoridis & Tsadiras, 2025).

1.4.2 Business Intelligence and Inventory Decision Support

Business Intelligence systems are central to the process of transforming predictive analytics into operational inventory decisions by building a single decision-support environment that incorporates data storage, analytical models, and managerial reporting. Instead of considering forecasting as a specific analytical process, BI frameworks incorporate the predictive output into the dashboards, alerts, and optimisation processes shaping the procurement, replenishment, and stock allocation decision-making in day-to-day business activities. This integration enables managers to move beyond descriptive reporting and plan inventory by taking into account the anticipated demand conditions.

Empirical research demonstrates that BI-assisted forecasting is more effective in enhancing demand visibility and decreasing typical inventory wastes like overstocking and stockouts. The BI systems established based on the past sales data and regression-based forecasting

systems allow decision-makers to predict the demand trends of the products and respond to them by modifying stock levels to enhance responsiveness at the store level and regional level (Fauzan & Efgivia, 2025). Once the predictive outputs are integrated into BI dashboards, the forecast results are readable and actable and not abstract numerical estimates.

Beyond simple forecasting, BI systems enhance inventory decision-making by supporting optimisation and scenario evaluation. Inventory planning systems integrating the demand forecast and optimisation models allow companies to incorporate the right reorder levels and stock levels within cost and capacity limitation. Supply chain research evidence demonstrates that a combination of predictive analytics and inventory optimisation has been shown to improve inventory turnover and reduce holding costs without significantly affecting service levels (Maharana et al., 2025). These results point to the fact that the effectiveness of forecasting alone is not a complete picture, the business value of forecasts is revealed when they are operationalised using decision rules and planning logic.

Recent studies also prove the fact that more advanced analytics, which are incorporated into BI environments, are capable of forecasting and identifying potential stockout risks at an early stage. The target group of the stockout prediction models (ML-based) implemented into the decision-support systems enables managers to recognize high-risk items and instigate preventive replenishment measures. Such predictive monitoring has been shown to reduce service disruptions and improve inventory reliability, particularly in data-intensive retail environments (Liu et al., 2025).This supports the purpose of BI as an early-warning data processing metric more than a retrospective data reporting measure.

The communication of insights to the decision-makers is also closely associated with the effectiveness of BI in managing inventory. Literature on BI dashboard design states that the dashboards can transform into actual decision-support tools when they are consistently developed. Properly designed dashboards can give managers an immediate insight into the implication of forecasts, compare possible actions, and find underlying causes of inventory inefficiencies, which can minimize decision latency (Tadakala et al., 2025). This shows that visualization is not simply cosmetic but it is essential in transforming the outputs of analysis into business actions.

The changes in organisations that are moving towards data-driven decision-making further highlight the significance of BI-supported inventory systems. It is found that decision-support systems comprising of forecasting, policy simulation, and performance monitoring have helped the firms redefine inventory management policies as they grow analytically. These systems facilitate cohesive decision-making within the departments and decrease the use of ad hoc judgement resulting in more regular and transparent practices in inventory control (Sá et al., 2025)

Operationally, BI implementations in logistics and inventory processes are frequently reported to have led to a faster decision, less errors and better planning. BI systems improve data integration and reporting, making them automated, thereby decreasing manual processing and providing close to real-time stock readings, replenishment expenses, and supplier performance. This ability enables the use of inventory decisions based on real

demand indicators instead of timely periodic reports, which leads to quantifiable efficiency improvements (Salvador-Callalli & Andrade-Arenas, 2023).

Further developed BI architectures go even further and implement AI-based forecasting alongside optimisation and real-time feedback of data. The use of hybrid systems where predictive models are used with optimisation logic indicates that the results of the forecast can be used to directly determine the quantities of procurement, to make decisions on whether to store or not to store, and when to replenish. It has been demonstrated that such systems have been shown to lower overall inventory costs and enhance service levels, which is how BI can serve as an intermediate between predictive and prescriptive decision-making (Kumar & Goyal, 2025).

All in all, this literature gives a consistent signal that Business Intelligence systems create value not by simply coming up with forecasts but by integrating the forecasts into decision-support frameworks that inform inventory actions. The combination of BI dashboards, optimisation modules and decision rules allows organisations to ensure consistency between inventory planning and the anticipated demand, which in turn transforms the accuracy of the analytics into practical operational results. This literature substantiates the perception that BI has become an essential enabler of inventory decision support in the contemporary retail and supply chain setting.

1.4.3 Product-Level Data Analysis and Customer Behaviour

The literature of retail analytics consistently points towards the significance of granular sales data to figure out the customer buying behaviour and even to enhance predictive performance. Transaction and product level data capture finer details about individual buying behavior, product characteristics and category associations that cannot be seen when sales are compiled at higher levels. The aggregated sales numbers have a propensity to level out fluctuations in different products and across product lines, which may obscure significant behavioural tendencies like product substitution, complementary buying, and sensitivity to promotion (Fildes et al., 2022).

Both product and category level variables are demonstrated to be at the center stage in demand development and sales prediction. Retail demand is not just a temporal or geographic phenomenon, but product characteristics, as well as, category arrangements are clearly influential. Models with SKU-level and category-specific information are generally more accurate predictors than models using aggregate or store-level data, especially in short-term operational decisions, including inventory planning and replenishment (Fildes et al., 2022). This shows that product level granularity is necessary in order to capture the heterogeneity of demand among various items.

The academic literature also reveals that demand interactions often happen both intra and inter-category. Price promotion in one product or category may have a spill over effect that may affect the demand of categories that are related either by substitution or by buying complements. These cross-category demand effects suggest that the purchasing decisions of customers are not made at an individual product level, but instead at a category level and sub-

category level, which supports the significance of category and sub-category variables in retail analytics (Leeflang & Parreno-Selva, 2011).

On a more detailed level, it is observed that adding both intra-category and inter-category promotional data significantly improves the accuracy of the forecast. Retail datasets with high levels of dimensionality, containing SKU-level promotional variables, are able to capture demand shifts that are not directly observed by the more basic models. Experimental data reveals that intra-category promotional effects explain a significant percentage of changes in the forecast, and the product in the same category has a strong impact on the demand pattern of other products (Ma et al., 2016). This justifies the application of product hierarchy data like category and sub-category in predictive modelling.

Product level analysis is also supported by the fact that there are differences in customer behaviour among products. Retail products differ in regards to the purchasing frequency, sensitivity of price and stability of demand. The high-frequency, low-value products tend to be more likely to follow a pattern of habitual buying and have stable demand, whereas high-value or low-frequency products tend to be more volatile and price and promotion-sensitive. Analytical models which do not differentiate the two types of products run the danger of simplifying customer behaviour and delivering less dependable forecasts (McCardle et al., 2007).

Strategically, product-level performance analysis is also vital in retail decision-making in regard to promotions, bundling and assortment planning. In that respect, bundling strategies are particularly dependent on the knowledge of the behavior of various products both separately and together. It has also been shown that the profitability and performance of bundles are based on the nature of demand of the products being packed together such as whether they are stapled goods or discretionary goods hence further emphasising the significance of breakdown of the product attributes as opposed to aggregated sales figures (McCardle et al., 2007).

The latest trends in machine learning and data analytics support those findings. Research that uses machine learning methods on supermarket transaction data shows that at the product level, the distinctiveness of specific purchasing behaviors, segments of customers, and co-purchase relationships can be identified. These insights aid in better targeting, personalised promotions and more accurate demand forecasting as opposed to methods that operate on rough-grained variables (Stylianou & Pantelidou, 2025). Likewise, predictive modelling, used with customer segmentation, demonstrates product-focused purchasing behaviour as one of the most important inputs to customer group classification and future demand forecasting (Tang, 2025).

In general, the literature gives solid empirical evidence on the inclusion of the variables of product category and sub-category in retail analytics. In a retailing context that involves a high number of transactions, product-level attributes are always more effective in explaining the variation in demand than either geography or aggregate variables. This is exactly what the analytical design of the present project includes category and sub-category variables in the dataset, and the results of the importance of features show that product characteristics are

more significant than geographic variables in predicting the outcome of retail sales (Fildes et al., 2022; Ma et al., 2016).

1.4.4 Comparison of Machine Learning Techniques for Sales Forecasting

Retail sales forecasting is typified by a complicated demand pattern due to seasonality, promotions, product diversification as well as customer behaviour. These can cause non-linear relationships and high dimensionality of sales datasets that the traditional statistical forecasting methods including linear regression and classical time-series models often face limitations when applied to high-dimensional and non-linear retail sales data. With the expansion of the volume and intricacy of retail datasets, sophisticated machine learning approaches have become utilized to ensure a superior reflection of these patterns and enhance predictive accuracy (Saptadi et al., 2025).

Recent literature consistently reports that ensemble machine learning models tend to outperform single predictive models in retail sales forecasting tasks when it comes to the retail sales forecasting tasks. Ensemble methods use a combination of decision trees to decrease variance, enhance generalisation, and strength against noise in transactional data. Comparative studies across retail environments show that ensemble models achieve lower prediction errors than single decision trees and linear models, particularly when datasets involve multiple explanatory variables such as product attributes and temporal effects (Ismail et al., 2025; Kang, 2023).

Random Forest has been extensively implemented in the field of retail sales prediction because of its strength and the capacity to reduce overfitting by means of bootstrap aggregation and random features selection. Empirical analyses show that Random Forest works well in various retail settings, and it demonstrates stable predictive performance with relatively lower sensitivity to hyperparameter tuning compared to boosting-based models. In retail datasets where interpretability and computational efficiency are important, Random Forest provides a practical balance between predictive performance and operational feasibility (Ganguly & Mukherjee, 2024; Xu et al., 2025).

Gradient Boosting models also add value to predictive accuracy in that a sequential learning approach is used as each subsequent model is meant to correct the previous model error. This iterative refinement enables Gradient Boosting to model complex non-linear interactions more precisely. Research on the retail setting comparing the performance of Random Forest with Gradient Boosting states that in the retail setting with changing demand behavior and varying product types, Gradient Boosting frequently achieves slightly lower error metrics (RMSE and MAE) in datasets characterised by complex non-linear demand patterns. (Kang, 2023; Kizgin et al., 2025).Nonetheless, there is also a trade-off in the predictive accuracy and the computational complexity as pointed out in the literature. Gradient Boosting often achieves higher forecasting accuracy but typically requires more extensive hyperparameter tuning and greater computational resources. Random Forest models, conversely, are simple to train and deploy, and are therefore frequently used as baseline or benchmarking models in applied retail analytics studies. (Ganguly & Mukherjee, 2024; Saptadi et al., 2025).

The evidence provided by systematic reviews of machine learning implementation in sales prediction also shows that there is no universal model that can dominate in every retail setting. Instead, the ensemble techniques like the Random Forest and Gradient Boosting are more successful when considered in the comparative mode that will enable the practitioners to compromise accuracy, interpretability, and operational limits (Saptadi et al., 2025; Ismail et al., 2025).

All in all, both Random Forest and Gradient Boosting have been chosen by the literature as suitable tools in retail sales forecasting. The combination of both models allows a more detailed assessment of performance trade-offs with the strengths of Random Forest being robustness and interpretability, and Gradient Boosting being more predictive on complex non-linear retail data. This type of comparison is consistent with standard research methods and offers a reasonable foundation to the final model choice on the basis of empirical performance measures like Average Squared Error (ASE) (Kang, 2023; Xu et al., 2025).

2.0 Data Preparation and Exploration

2.1 Data Cleaning and Preprocessing

The Superstore-Sales-Dataset was originally in an Attribute-Relation File Format (ARFF) and was opened as a text file to examine its structure before being converted into a Comma-Separated File. The dataset was then loaded into SAS Enterprise Guide for data cleaning and preprocessing. Firstly, 11 missing values were identified in the Postal_Code column. These records were filtered out using the Query Builder function by applying the condition Where "Postal_Code" NOT = '?'. Besides, a box plot analysis was performed to examine the distribution of sales variable and determine significant outliers. Tukey's Fence method was used to confirm the findings and the results indicated a significant right skew with a maximum value of $22638 compared to a 99th percentile of $2480. To mitigate the influence of these extreme values, 99th-percentile winsorization was applied to using Query Builder. This preprocessing step ensures that the results of the random forest and gradient boosting model generalize effectively.

2.2 Exploratory Data Analysis

Summary Statistics Results

Analysis Variable : Sales_Capped					
Mean	Std Dev	Minimum	Maximum	N	Median
209.4932887	394.2419921	0.4440000	2480.00	9789	54.3840000

Table 2.2.1 Summary Statistics for Sales_Capped

Descriptive statistics were generated for the Sales_Capped variable to examine central tendency and dispersion. The analysis reveals a mean sales value of 209.49, which is significantly higher than the median of 54.38. This disparity indicates a highly right-skewed

distribution. The standard deviation of 394.24 reflects substantial variability, confirming that while most transactions are low value, the revenue is heavily influenced by high-value orders.

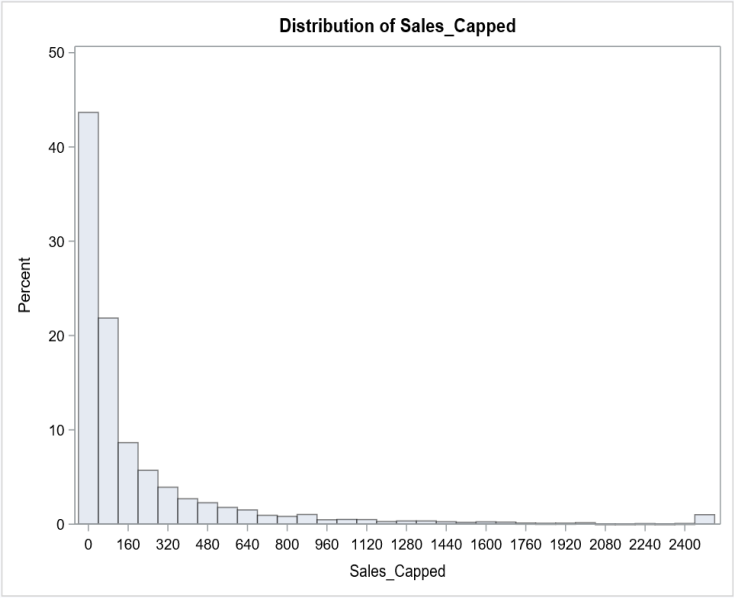


Figure 2.2.2 Histogram of Sales_Capped Distribution

The histogram confirms the positive skew of the sales data, with the highest frequency of orders clusters below the 160 marks, with frequency rapidly declining as sales values increase. There is a long tail extending toward the maximum value of 2480.00.

One-Way Frequencies
Results

Segment	Frequency	Percent	Cumulative Frequency	Cumulative Percent
Consumer	5096	52.06	5096	52.06
Corporate	2948	30.12	8044	82.17
Home Office	1745	17.83	9789	100.00

Table 2.2.3 Frequency Result of Customer Segment

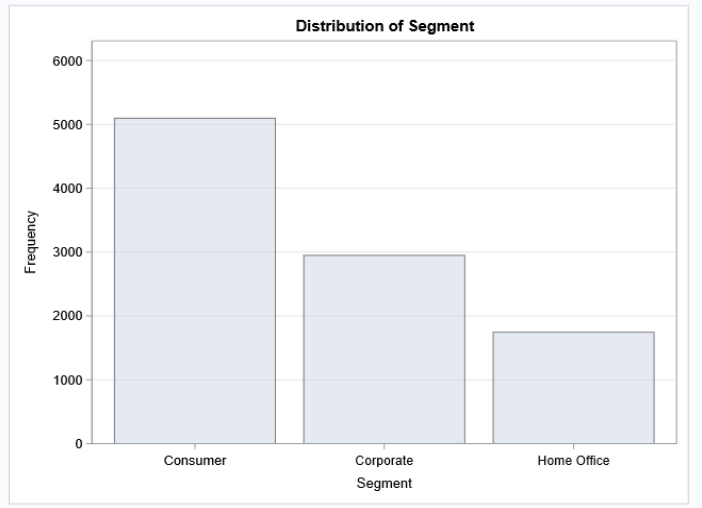


Figure 2.2.4 Frequency Distribution of Customer Segments

Table 2.2.3 and Figure 2.2.4 details the breakdown of orders by customer segment, showing that the Consumer segment is the dominant contributor with 5,096 transactions and accounts for 52.06% of the total. The Corporate segment has 2,948 transactions (30.12%) while the Home Office segment represents the smallest portion of the dataset with 1,745 transactions (17.83%).

Ship_Mode	Frequency	Percent	Cumulative Frequency	Cumulative Percent
First Class	1501	15.33	1501	15.33
Same Day	538	5.50	2039	20.83
Second Class	1901	19.42	3940	40.25
Standard Class	5849	59.75	9789	100.00

Table 2.2.5 Frequency Result of Ship Modes

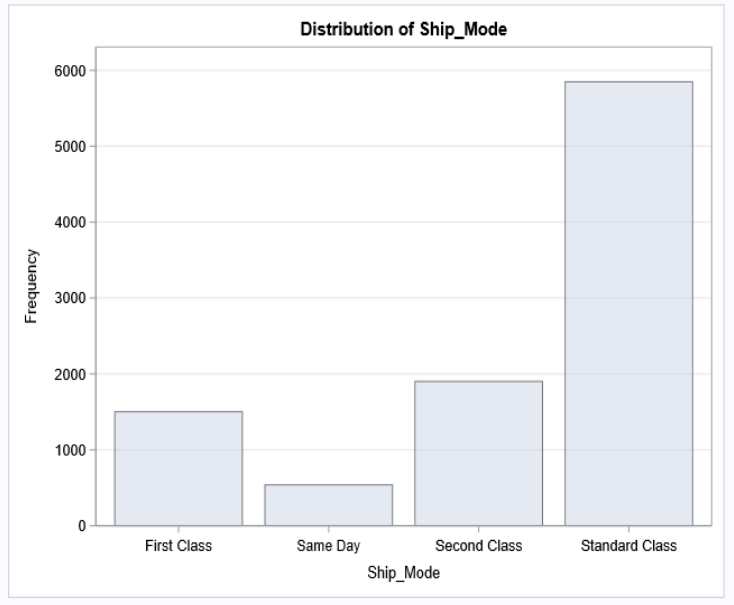


Figure 2.2.6 Frequency Distribution of Ship Modes

Table 2.2.5 and Figure 2.2.6 categorise the shipping methods chosen by customers. It was revealed that Standard Class is the most preferred option with 5,849 transactions and was 59.75% of the total. Second Class and First Class followed with 19.42% and 15.33% of the shipping modes respectively, while Same Day shipping is the least utilized, accounting for only 5.50% of the total orders.

Region	Frequency	Percent	Cumulative Frequency	Cumulative Percent
Central	2277	23.26	2277	23.26
East	2774	28.34	5051	51.60
South	1598	16.32	6649	67.92
West	3140	32.08	9789	100.00

Table 2.2.7 Frequency Result of Region

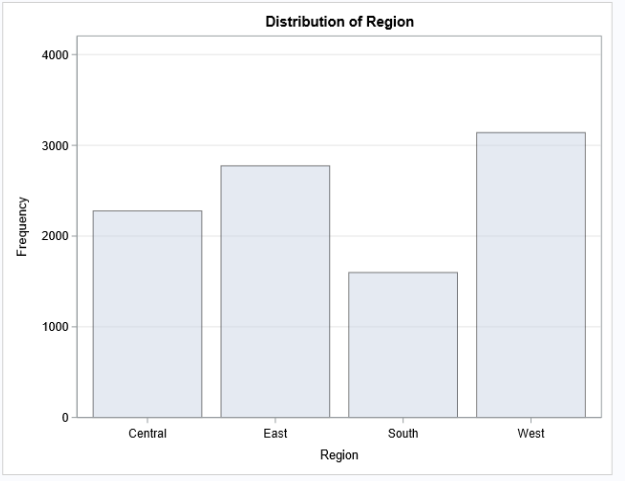


Figure 2.2.8 Frequency Distribution of Region

Table 2.2.7 and Figure 2.2.8 tell the transaction by geographic region. The West region was the most active region with 3,140 transactions (32.08%). The East region has 2,774 transactions (28.34%) and the Central region has 2,277 transactions (23.26%). The South region has the lowest sales activity with 1,598 transactions and was 16.32% of the total sales.

Category	Frequency	Percent	Cumulative Frequency	Cumulative Percent
Furniture	2076	21.21	2076	21.21
Office Supplies	5903	60.30	7979	81.51
Technology	1810	18.49	9789	100.00

Table 2.2.9 Frequency Result of Category

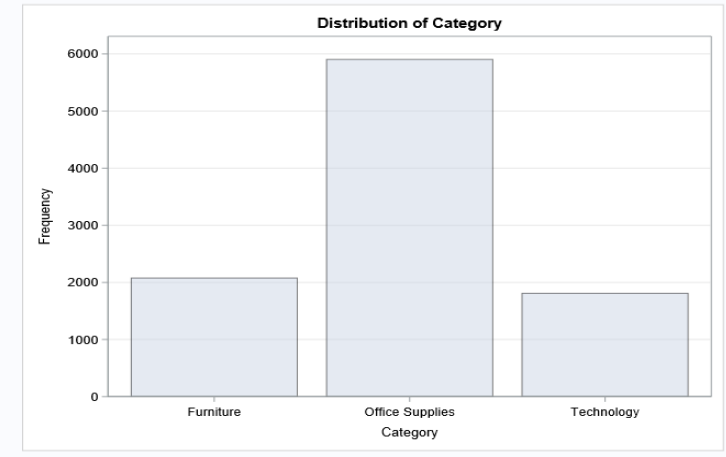


Figure 2.2.10 Frequency Distribution of Category

Table 2.2.9 and Figure 2.2.10 shows the frequency of orders by product category. It revealed that Office Supplies is the most popular category with 5,903 transactions and was 60.30% of the entire transaction. Furniture and Technology have much lower frequencies and represent 21.21% and 18.49% of transactions respectively. This distribution highlights that while the store sells products of three categories, the volume of transactions is heavily concentrated in low-value, high-frequency office supply items rather than furniture or technology.

Sub-Category	Frequency	Percent	Cumulative Frequency	Cumulative Percent
Accessories	754	7.70	754	7.70
Appliances	458	4.68	1212	12.38
Art	784	8.01	1996	20.39
Binders	1492	15.24	3488	35.63
Bookcases	225	2.30	3713	37.93
Chairs	606	6.19	4319	44.12
Copiers	66	0.67	4385	44.80
Envelopes	247	2.52	4632	47.32
Fasteners	214	2.19	4846	49.50
Furnishings	931	9.51	5777	59.02
Labels	357	3.65	6134	62.66
Machines	115	1.17	6249	63.84
Paper	1336	13.65	7585	77.48
Phones	875	8.94	8460	86.42
Storage	831	8.49	9291	94.91
Supplies	184	1.88	9475	96.79
Tables	314	3.21	9789	100.00

Table 2.2.11 Frequency Result of Subcategory

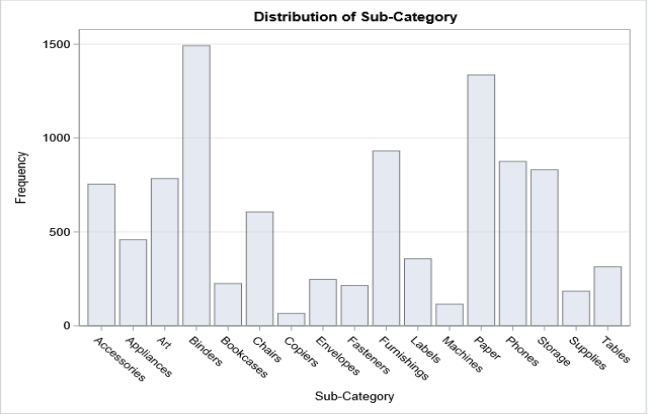


Figure 2.2.12 Frequency Distribution of Subcategory

Table 2.2.11 and Figure 2.2.12 lists the transaction counts for specific sub-categories. The subcategories with the highest sales are Binders (15.24%) and Paper (13.65%). Conversely, higher value items like Copiers (0.67%) and Machines (1.17%) have the lowest transaction counts, indicating that they are purchased far less frequently than general supplies.

2.3 Visualization

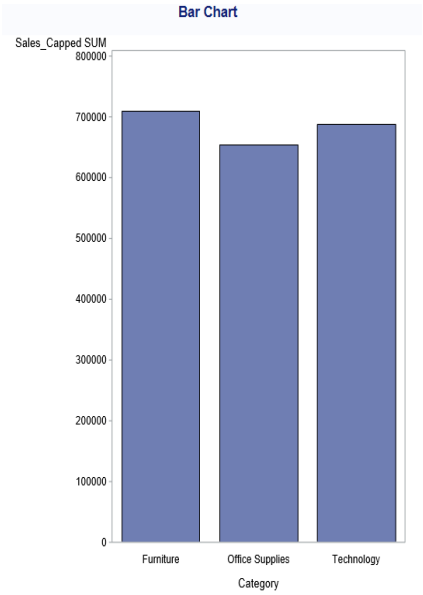


Figure 2.3.1 Total Sales Revenue by Product Category

The figure above compares total sales revenue across product categories. It was revealed that the Furniture category generates the highest total revenue, exceeding the 700,000 mark. On the other hand, while Office Supplies has the highest transaction volume, it generates the lowest total revenue as compared to the other categories likely due to lower unit prices.

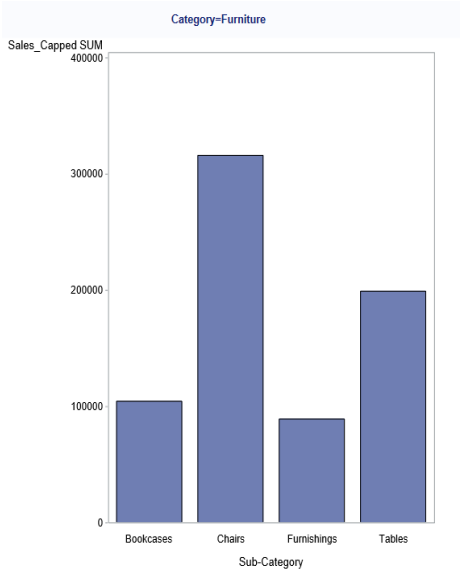


Figure 2.3.2 Total Sales Revenue by Furniture Subcategory

The bar chart above breaks down the total sales revenue within the Furniture category. It was revealed that Chairs was the dominant revenue generator with sales exceeding 300,000. Tables rank as the second highest contributor reaching approximately 200,000 in revenue. On the other hand, Bookcases and Furnishings generate significant lower revenue with both generating nearly 100,000 in revenue.

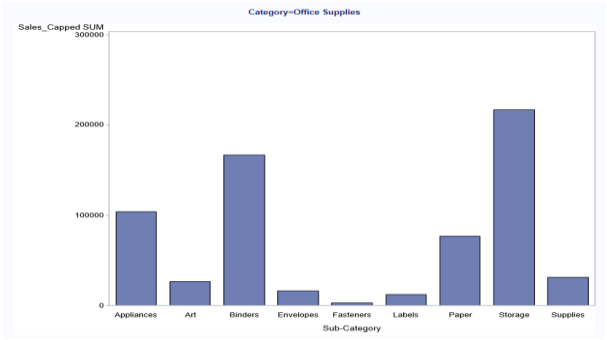


Figure 2.3.3 Total Sales Revenue by Office Supplies Subcategory

The bar chart above displays the sum of sales for each Office Supplies sub-category. Storage records the highest total sales, exceeding 200,000, followed by Binders, which also performs strongly with over 150,000 in total sales. Conversely, Fasteners, Labels, and Envelopes show minimal revenue impact compared to the category leaders.

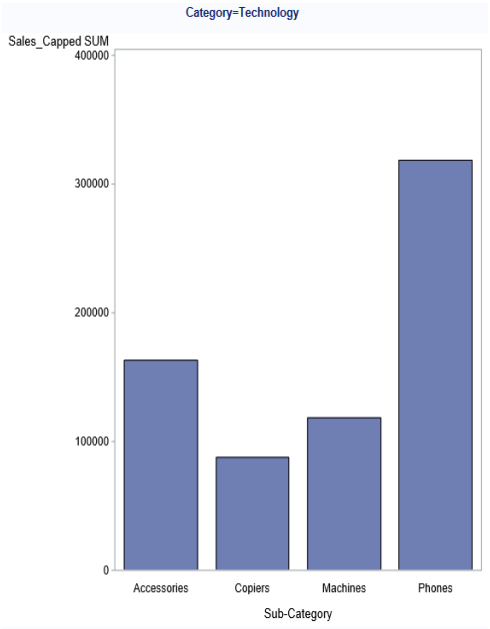


Figure 2.3.4 Total Sales Revenue by Technology Subcategory

The figure above highlights the Phone category having the most sales surpassing 300,000 in total sales. Followed by the Accessories category being the second largest revenue source. Machines and Copiers have the lowest generated revenues for the Technology sector with Copiers recording sales under 100,000.

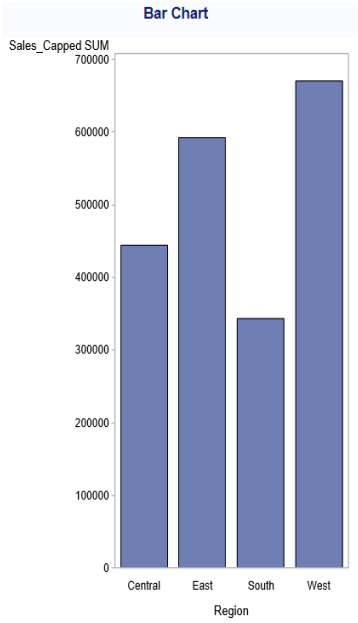


Figure 2.3.5 Regional Sales Performance

The figure illustrates total sales revenue by region, with the West region having the strongest sales performance of 700,000. The East region was the second highest performer followed by the Central region. The South region generates the lowest revenue below the 400,000 mark.

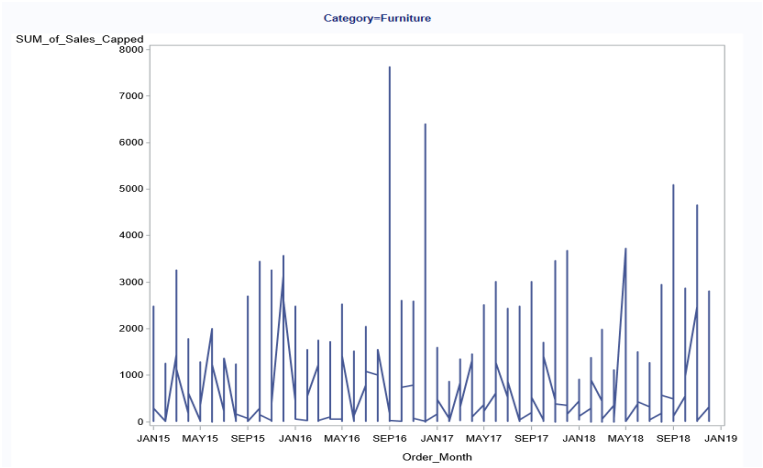


Figure 2.3.6 Monthly Sales Trends for Furniture

The time-series line chart for Furniture displays high volatility in sales revenue over the four-year period, suggesting that furniture purchases are irregular and likely influenced by seasonal demand. Notable peaks occur frequently, with the highest spike exceeding 7,000 in revenue.

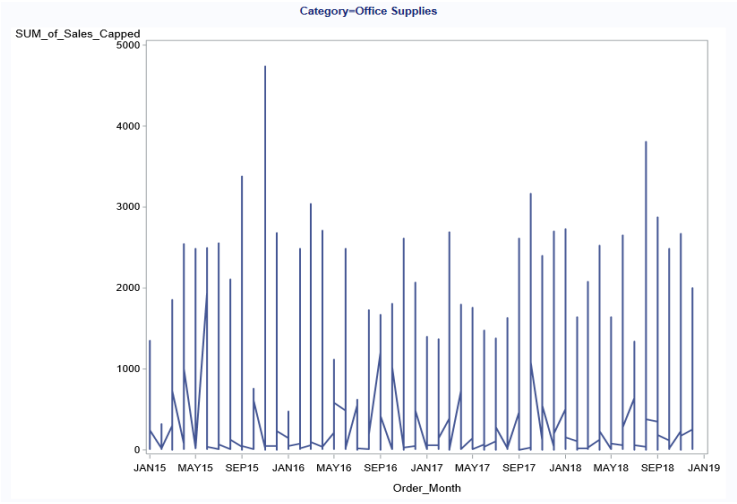


Figure 2.3.7 Monthly Sales Trends for Office Supplies

The figure above shows the trend line for Office Supplies and shows a dense pattern of fluctuations, which reflects the high demand in this category. While it experiences frequent spikes, the maximum revenue peaks are generally lower than those in Furniture or Technology categories, mostly staying below 4,000, with one exceptional spike reaching approximately 4,800 sales.

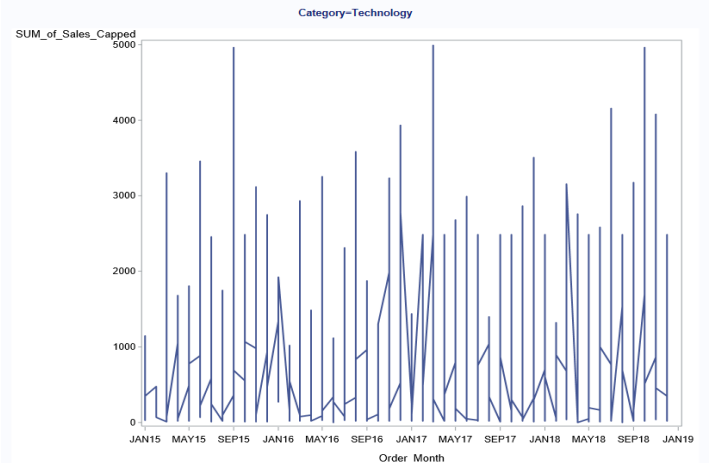


Figure 2.3.8 Monthly Sales Trends for Technology

The figure above tracks the sales performance for Technology products. The time-series was characterised by distinct, sharp peaks that frequently reached 4,000 to 5,000 in sales. This reflects that technology sales are driven by high-value purchases rather than a consistent flow of daily revenue.

3.0 Predictive Modelling

3.1 Model Technique 1 – Random Forest

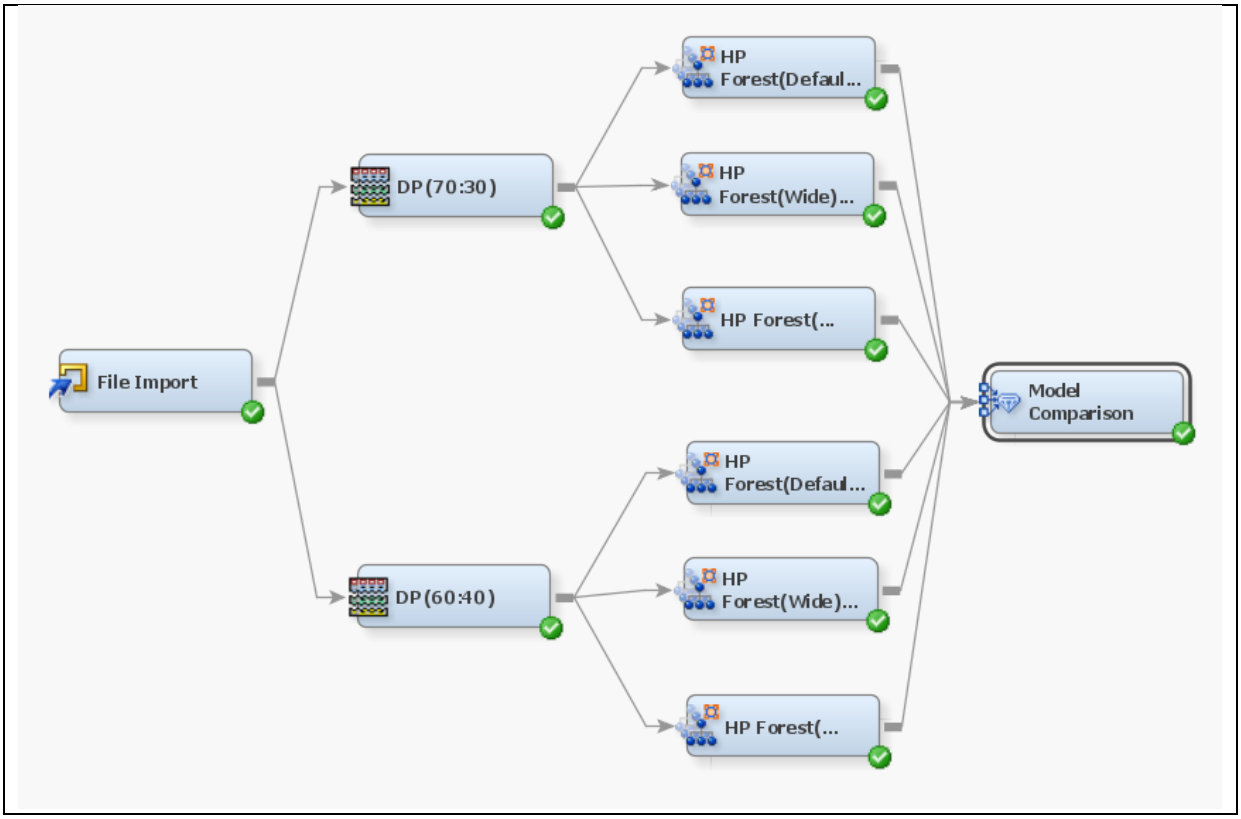
Random forests are defined as a combination of tree predictors, where each tree is influenced by the values of a random vector which been sampled independently and maintain a same distribution across all trees in the forest (Breiman, 2001). This model will be implemented in our study via Sas Enterprise Miner.

Multiple Random Forest models were implemented by systematically adjusting the key model parameters. In this study, we adjust the maximum number of trees, method of leaf size and different data partition strategies. To evaluate model performance, Average Squared Error (ASE) is used as the primary criterion for model selection as it is suitable for continuous target prediction and can ensure generalisation and prevent overfitting.

The following table and figure show the 6 different random forest model with different characteristics.

Node	**Name**	**Data Partition (Train: Validation)**	**Parameter Adjustment**
HP Forest (Default) M1	RF (No Change)	70:30	No adjustment
HP Forest (Wide) M2	RF (Maximum Number of Trees =200)		Number of trees adjust from 100 to 200
HP Forest (Conservative) M3	RF (Method of Leaf Size = Count)		Method of Leaf Size adjust from Default to Count
HP Forest (Default) M4	RF (No Change)	60:40	No adjustment
HP Forest (Wide) M5	RF (Maximum Number of Trees =200)		Number of trees adjust from 100 to 200
HP Forest (Conservative) M6	RF (Method of Leaf Size = Count)		Method of Leaf Size adjust from Default to Count

SAS Miner Diagram



Fit Statistics using Model Comparison

Fit Statistics

Model Selection based on Valid: Average Squared Error (_VASE_)

Selected Model	Model Node	Model Description	Valid: Average Squared Error	Train: Average Squared Error
Y	HPDMForest	HP Forest(Default)M1	115571.48	119098.50
	HPDMForest2	HP Forest(Default) M4	115623.12	120821.47
	HPDMForest6	HP Forest(Conservative) M6	115623.12	120821.47
	HPDMForest4	HP Forest(Conservative) M3	115812.72	119278.18
	HPDMForest3	HP Forest(Wide) M2	115815.04	119546.07
	HPDMForest5	HP Forest(Wide) M5	115827.74	120942.50

Model selection was based on validation Average Squared Error (ASE). As shown in the fit statistics above, the model HPDM Forest achieved the lowest validation ASE (115,571.48). This model will be selected as the best performance model as it has the lowest validation ASE.

3.2 Model Technique 2 – Gradient Boosting

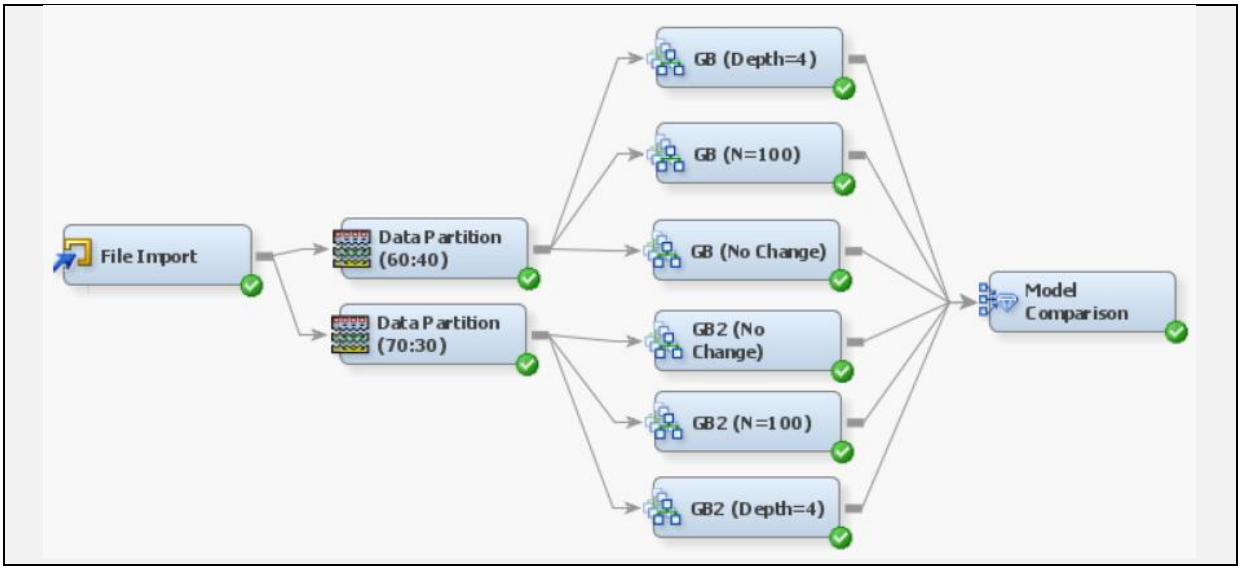
Gradient boosting is an ensemble learning algorithm that provide accurate predictions by combining multiple decision trees into a single model (Clark & Lee, 2025). We will use Gradient Boosting node in SAS Enterprise Miner to perform the model.

Multiple Gradient Boosting models were developed by systematically adjusting key model parameters. In this study, we adjust the number of boosting iterations, maximum tree depth and different data partition strategies. Model performance was evaluated using Average Squared Error (ASE), which is appropriate for continuous target prediction. The validation dataset was used as the primary criterion for model selection to ensure generalisation and prevent overfitting.

The following table and figure show the 6 different gradient boosting model with different characteristics.

Node	**Name**	**Data Partition (Train: Validation)**	**Parameter Adjustment**
Boost	GB(No Change)	60:40	No adjustment
Boost5	GB (N=100)		Number of iterations adjust from 50 to 100
Boost6	GB(Depth = 4)		Number of maximum depths adjust from 2 to 4
Boost2	GB2 (No Change)	70:30	No adjustment
Boost3	GB2 (N=100)		Number of iterations adjust from 50 to 100
Boost4	GB2(Depth = 4)		Number of maximum depths adjust from 2 to 4

SAS Miner Diagram



Fit Statistics using Model Comparison

Fit Statistics
Model Selection based on Valid: Average Squared Error (_VASE_)

Selected Model	Model Node	Model Description	Valid: Average Squared Error	Train: Average Squared Error
Y	Boost	GB (No Change)	111954.34	107076.35
	Boost5	GB (N=100)	111954.34	107076.35
	Boost4	GB2 (Depth=4)	111977.55	97991.31
	Boost6	GB (Depth=4)	112161.88	97523.15
	Boost2	GB2 (No Change)	112528.95	106772.31
	Boost3	GB2 (N=100)	112528.95	106772.31

Model selection was based on validation Average Squared Error (ASE). As shown in the fit statistics above, the model Boost and Boost 5 achieved the lowest validation ASE (111,954.34). Although both models have the same ASE, but we will choose Boost as the best performance model as it has less complexity compared to Boost5. Therefore, Boost was chosen as the final model.

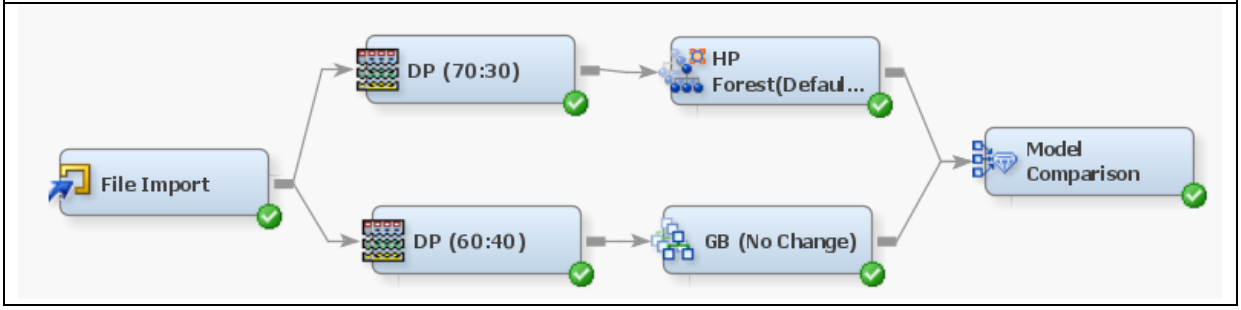
3.3 Model Comparison

The best model from Random Forest and Gradient Boosting will be selected to compared with each other. **Average Squared Error (ASE)** will be the primary metric to accurately determine the predictive performance of the models, as it provides a direct

measurement between the predicted and actual sales figures. A lower ASE indicates superior model accuracy.

Model	ASE	Complexity
Random Forest	115,571.48	Low (0.90 seconds)
Gradient Boosting	111,954.34	High (4.53 seconds)

SAS Miner Diagram



Fit Statistics using Model Comparison

Fit Statistics
Model Selection based on Valid: Average Squared Error (_VASE_)

Selected Model	Model Node	Model Description	Valid: Average Squared Error	Train: Average Squared Error
Y	Boost	GB (No Change)	111954.34	107076.35
	HPDMForest7	HP Forest(Default)M1	115508.79	119313.57

Based on a comprehensive evaluation, **Gradient Boosting** is selected to be the optimal model as it outperformed Random Forest model across key performance metrics.

On accuracy, Gradient Boosting model outperformed Random Forest model on the validation set with ASE of 111,954.34 compared to Random Forest model of 115,508.79. This represents a performance improvement of approximately 3.1%, indicating that Gradient Boosting model able to capture the complex patterns in the sales data more effectively.

On generalization and overfitting, Gradient Boosting model also shows a healthy generalization pattern with Training ASE (107,076.35), which this small gap can confirm that the model has not significantly overfitted and in a well-fitted model. Meanwhile, The Random Forest model exhibited a slightly unusual pattern where the Training ASE

(119,313.57) having higher value than Validation ASE (115508.79), suggesting that the model has slightly underfitting.

In terms of computational cost, Random Forest was more effective as it requires only 0.90 seconds to train due to its parallel processing structure. While Gradient Boosting was more complex that takes **4.53 seconds,** approximately 5 times longer than Random Forest due to its sequential learning approach where each tree corrects the previous one.

With this comparison, we ultimately decided on using Gradient Boosting model as although the Gradient Boosting model incurs a higher computational cost, the trade-off is fully justified. It demonstrates superior accuracy of 3.1% gain and better generalization on unseen data. Therefore, we accept the higher complexity of Gradient Boosting to achieve reliability and precision in our sales forecasting.

3.4 Feature Importance

Following the selection of Gradient Boosting as the optimal model, we analysed its **Variable Importance** matrix to identify the primary drivers of sales. The importance values are scaled relative to the most influential variable (1.0).

Variable Importance

Variable Importance

Variable Name	Label	Number of Splitting Rules	Importance	Validation Importance	Ratio of Validation to Training Importance
Sub_Category		48	1	1	1
Postal_Code	Postal_Code	27	0.467005	0	0
Category	Category	4	0.265906	0.2186	0.822095
City	City	2	0.149157	0	0
Region	Region	0	0	0	.
Segment	Segment	0	0	0	.
State	State	0	0	0	.
Country	Country	0	0	0	.
Ship_Mode	Ship_Mode	0	0	0	.

Variable importance analysis was used to identify the produce characteristics as primary driver that affect sales performance. Specifically, Sub_Category emerged as the the most influential variable with importance score of 1.0, indicating that the specific type of product sold have a major significant on sales performance.

Another important predictor is Category with important score of approximately 0.27 and maintain a meaningful predictive power on validation important score of approximately 0.22. This means that product category presents a reliable predictor for predicting future sales data. Besides that, Postal_Code (~0.47) and City (~0.15) appear to be important predictors during training phrase, but their important scores dropped to 0 when applied to the validation data. This indicates that the model is able to memorize specific high-performing locations within the training dataset but failed to generalize this pattern into the new data. This finding

highlights a strong evidence that sales performance is not essentially location-dependent, as a zip code that performed well in the past does not guarantee the same predictor for the future.

Consequently, the predictive power of Gradient Boosting model is almost determined by **what** is being sold (Sub_Category) rather than **where** it is sold (City, Region, State). The use of geographic variables as a predictor of unseen data was found to be insignificant.

4.0 Business Insights and Recommendations

This session will discuss how businesses can understand the findings of the predictive model and transform them into actual actions and strategies. It explains how businesses can correlate the results of a predictive model with the daily operation of businesses. For example, create better data-driven sales forecasting and reallocation of resources to enhance the operational efficiency and competitive performance in the marketplace.

4.1 Model Interpretation

4.1.1 Reliability of Sales Forecasts for Business Decision-Making

This project has chosen Gradient Boosting as the final predictive model that may provide a good basis and resource for short-term sales prediction. From a business perspective, the most important point is the confidence that Gradient Boosting can provide to company management in terms of predicting future sales patterns, rather than being the best Year 2 model. Since this model is able to generalize to future unseen data, it may help company management to make daily operational decisions accurately based on the model's prediction results, including inventory restock, purchasing planning, and sales targeting. Therefore, this model may help to reduce the instability of planning and encourage the transformation of a company to a more proactive and data-driven one. The result also states the ability to enable management to predict demand changes earlier and adopt suitable strategies on time to avoid certain risks, including overstocking and missed sales opportunities.

4.1.2 Product-Level Drivers of Sales Performance

The observations of the predictive model in this project show that the main driver of sales performance is product characteristics, instead of geographic or external market conditions. Besides, the analysis of feature importance mentioned above also highlights that the Sub_Category product type is the most important factor that may affect sales performance. Hence, it can be concluded that specific product types have a significant effect on revenue generation.

Therefore, steps of identifying and focusing on high-performing Sub_Category is important since it will help to increase the overall sales performance from a sales perspective. In this case, products with high continuous demand should be given more attention while making various business decisions, including inventory plans, merchandising strategies, and

promotion strategies. Meanwhile, various strategies that may help to increase sales should be adopted for products that have low demand, including rationalization, repositioning, or promotional strategies. Overall, these product-based insights may help the company to gain advantages and increase the return value by allocating human, financial, and physical resources effectively.

4.1.3 Strategic Value of Category Level Insights

Besides Sub_Category, the category variable also provides important insights into predicting sales since it provides insights into macro level strategic decisions. Therefore, although it may not provide detailed information as the Sub_category variable, it also provides an overall view of the demand trends on products. These higher level views may enable businesses to make higher level strategic decisions. For example, make bundle sales for complementary products or promotion sales based on product categories.

For instance, if a specific product category contains stable high demand, the company may create a promotional campaign or bundle sales to encourage customer demand and sales performance. Besides, the category insights from the predictive model also help the company to decide merchandising decisions, including which product should have more shelf space, and which product should receive more visual presentations compared to others. Thus, the category level insight from the result is important to decide and structure the overall business strategy, while Sub_Category insights will help more with operational actions.

4.1.4 Limited Business Impact of Geographic Factors

Location may have led to good historical sales performance in other cases. However, the observed result of this project shows a limited influence of land-based variables on sales performance, including city and postal code. Based on the predictive model result, there is little or no correlation between the historical performance of a particular place and its future sales results. Therefore, the stability of selling in a specific location is lower than the product factors from a business perspective.

As a result, there are specific and immediate managerial effects from these items. For example, if a company utilizes strategies that are based on location, including marketing and inventory management by location, it may not be able to produce a stable return on investment (ROI) continuously. Meanwhile, adoption of product-focused strategies to allocate resources and stock may have more influence on sales since the data shows a more stable and consistent correlation between product and sales performance. Hence, companies may focus on measurable performance improvement initiatives to reduce the operational complexity, including using product-focused management instead of location-based management.

4.1.5 Enabling Proactive and Data-Driven Business Strategy

The interpretation of the predictive model also shows the value of analytics in guiding businesses to adopt the best resource allocation strategies. Companies are not required to make decisions based on guessing and make practical strategies based on facts after acquiring accurate sales forecasts. Predictive models also provide chances to businesses to improve their inventory, marketing, and sales management by supplying insights related to products to management. Therefore, the process of purchasing decisions may be easier due to the reduced dependence of companies on geographic area.

Overall, businesses are allowed to plan their future better to allocate resources properly and enhance operational efficiency by turning predictive model results into practical business insights. Therefore, it also provides insights into recommendation sessions, which may help businesses increase sales performance and competitive power within the marketplace.

4.2 Recommendations

First and foremost, management is encouraged to focus on inventory management strategy by using an analytic methodology for the Sub_Category Level. The Gradient Boosting model should be used to calculate suitable inventory levels for high-demand items. Therefore, it may help businesses to increase sales performance and avoid certain risks that may reduce revenue. Meanwhile, for products that have low demand on the Sub_Category Level, strategies such as targeted promotion events or purchasing adjustments are required to reduce the holding costs and improve inventory turns.

Other than that, practical insights have been provided to marketing teams by analyzing the Sub_Category and Category-level feature importance. These may help them to target accurately on the Sub_Category that obtained high performance in promotional campaigns, advertisements, and social media efforts. In this case, businesses are encouraged to develop bundle sales, cross-selling programs, and seasonal promotions based on the insights of the category level, and increase total revenue by utilizing broader product trends. Therefore, management of businesses is able to allocate their resources efficiently and enhance return on investment (ROI) through the combination of marketing strategies and predictive insights.

Furthermore, the accurate sales prediction provided by the gradient boosting model has provided evidence and guidance to operational decision makers to make their decisions efficiently. While procurement and prediction demand for popular products can be synchronized, it may reduce the effect brought by overstocking and the related costs. Besides, management may also adjust the number of staff based on sales volume to maximize and enhance the workforce allocation. Therefore, it may help to enhance the efficiency and customer service quality. The predictive results may also help with investment decisions

regarding financial resources. It may help to maximize ROI by allocating financial resources efficiently to marketing, operational, and inventory management.

Moreover, regular updates on the latest sales data are also important for the gradient boosting model to maintain and enhance the accuracy and relevance for sales forecasting. Continuous retraining may help the model to adapt to new situations, including unseen patterns, changes of season, and new products. Therefore, the model may be able to predict accurately due to continuous development and enhancement of the model. Businesses are required to pay attention to the model update process since the updated model may provide flexibility and agility in response to any changes in businesses. Meanwhile, it may also help to maintain the predictive nature of the model and make it actionable.

In a nutshell, the predictive insights should be included in the overall strategic plan of the businesses. High performance of sub-categories requires focused attention while planning product development, expansion, and inventory investment. Meanwhile, the category level trends should serve as guidance on merchandising and promotional strategy. Besides, resources that are currently used on location-based strategies may be reallocated to product-based strategies since the model shows that geographic characteristics do not have a correlation with sales performance. Therefore, businesses are able to adopt predictive analysis to improve performance, efficiency, and revenue by including predictive insights in operational and strategic decision making routinely.

5.0 YouTube Link

https://youtu.be/qJ1tmNzgtXE

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Appendix

https://youtu.be/qJ1tmNzgtXE